

PBL Draft

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Library Downloads

```
library(tidyverse)
library(sf)
library(units)
library(readr)
library(purrr)
library(RColorBrewer)
library(data.table)
library(tmap)
library(viridis)
library(dplyr)
library(stringr)

library(tidycensus)
library(tigris)
options(tigris_use_cache = TRUE)
CRS.new <- st_crs("EPSG:3435")
```

We are limiting this analysis to the year 2025. No way to actually filter CTA Rail data before download.

Data Retrieval

```
crimes_raw <- st_read("crimes_2025.csv")
```

```
## Reading layer 'crimes_2025' from data source
##   'C:\Users\enigh\OneDrive\Desktop\SOSC 13220\Final Project\crimes_2025.csv'
##   using driver 'CSV'
```

```
cta_rail_stations <- st_read("cta_rail_stations.csv")
```

```
## Reading layer 'cta_rail_stations' from data source
##   'C:\Users\enigh\OneDrive\Desktop\SOSC 13220\Final Project\cta_rail_stations.csv'
##   using driver 'CSV'
```

```
police_station_raw <- st_read("police_stations.csv")
```

```
## Reading layer 'police_stations' from data source  
## 'C:\Users\enigh\OneDrive\Desktop\SOSC 13220\Final Project\police_stations.csv'  
## using driver 'CSV'
```

```
cta_ridership_raw <- st_read("cta_rail_ridership.csv")
```

```
## Reading layer 'cta_rail_ridership' from data source  
## 'C:\Users\enigh\OneDrive\Desktop\SOSC 13220\Final Project\cta_rail_ridership.csv'  
## using driver 'CSV'
```

Load in polygon dataset:

```
com_areas <- st_read("comm_area.geojson")
```

```
## Reading layer 'comm_area' from data source  
## 'C:\Users\enigh\OneDrive\Desktop\SOSC 13220\Final Project\comm_area.geojson'  
## using driver 'GeoJSON'  
## Simple feature collection with 77 features and 9 fields  
## Geometry type: MULTIPOLYGON  
## Dimension: XY  
## Bounding box: xmin: -87.94011 ymin: 41.64454 xmax: -87.52414 ymax: 42.02304  
## Geodetic CRS: WGS 84
```

```
chicago_union <- st_union(com_areas) # using this to get rid of CTA stations outside of Chicago
```

Data Cleaning/Wrangling

We need to filter CTA ridership data to just the year 2025. We also need to aggregate data, as ridership is currently broken down into months.

```
cta_ridership_clean <- cta_ridership_raw %>%  
  mutate(YEAR = as.integer(YEAR)) %>%  
  filter(YEAR == 2025)  
  
cta_ridership_clean <- cta_ridership_clean %>%  
  mutate(TOTAL_RIDES = as.numeric(TOTAL_RIDES)) %>% # ensure numeric  
  group_by(RIDERSHIP_ID, NAME) %>%  
  summarise(  
    total_rides_2025 = sum(TOTAL_RIDES, na.rm = TRUE),  
    .groups = "drop"  
  ) %>%  
  arrange(desc(total_rides_2025))  
  
head(cta_ridership_clean)
```

```
## # A tibble: 6 x 3  
##   RIDERSHIP_ID NAME          total_rides_2025
```

```
##      <chr>          <chr>          <dbl>
## 1 1660             Lake/State        3212890
## 2 890              O'Hare Airport    3005653
## 3 380              Clark/Lake        2798412
## 4 1220             Fullerton         2370481
## 5 1450             Chicago/State     2363911
## 6 260              State/Lake        2337971
```

```
head(crimes_raw)
```

```
##      ID Case.Number      Date      Block IUCR
## 1 14120097      JK160527 01/01/2026 12:00:00 AM 030XX W FRANKLIN BLVD 0810
## 2 14118983      JK159390 01/01/2026 12:00:00 AM 053XX S MICHIGAN AVE 1305
## 3 14112519      JK151612 01/01/2026 12:00:00 AM 102XX S EMERALD AVE 1565
## 4 14103838      JK140954 01/01/2026 12:00:00 AM 064XX S STEWART AVE 0810
## 5 14100237      JK136833 01/01/2026 12:00:00 AM 042XX W 31ST ST 0486
## 6 14096413      JK131515 01/01/2026 12:00:00 AM 006XX N LOCKWOOD AVE 1130
##      Primary.Type      Description Location.Description
## 1      THEFT              OVER $500      APARTMENT
## 2      CRIMINAL DAMAGE      CRIMINAL DEFACEMENT      OTHER (SPECIFY)
## 3      SEX OFFENSE INDECENT SOLICITATION OF A CHILD      RESIDENCE
## 4      THEFT              OVER $500      APARTMENT
## 5      BATTERY            DOMESTIC BATTERY SIMPLE      RESIDENCE
## 6      DECEPTIVE PRACTICE      FRAUD OR CONFIDENCE GAME      APARTMENT
##      Arrest Domestic Beat District Ward Community.Area FBI.Code X.Coordinate
## 1 false      false 1221      012      27      23      06
## 2 false      false 0225      002      3      40      14      1178064
## 3 false      false 2232      022      21      73      17      1173114
## 4 false      true 0722      007      6      68      06      1174731
## 5 false      true 1031      010      22      30      08B     1148654
## 6 false      false 1524      015      37      25      11      1140914
##      Y.Coordinate Year      Updated.On      Latitude      Longitude
## 1      2026 02/25/2026 03:43:24 PM
## 2      1869625 2026 02/25/2026 03:41:59 PM 41.797566426 -87.62254141
## 3      1837048 2026 02/18/2026 03:55:47 PM 41.708281913 -87.641654132
## 4      1862261 2026 02/09/2026 03:40:49 PM 41.777433807 -87.63498329
## 5      1883704 2026 02/05/2026 03:40:58 PM 41.836817925 -87.730030636
## 6      1903732 2026 02/05/2026 03:40:58 PM 41.891923124 -87.757939633
##      Location
## 1
## 2      (41.797566426, -87.62254141)
## 3      (41.708281913, -87.641654132)
## 4      (41.777433807, -87.63498329)
## 5      (41.836817925, -87.730030636)
## 6      (41.891923124, -87.757939633)
```

We are also going to ignore crimes entries that have no location data (as otherwise, it is not possible to gauge whether it occurred within a buffer or not.) We also see that the datasets have latitude and longitude data but not point. We will convert to point data as well.

Crimes:

```

crimes_clean <- crimes_raw %>% select(-Location)

crimes_clean <- crimes_clean %>%
  mutate(Latitude = as.numeric(Latitude), Longitude = as.numeric(Longitude))

crimes_clean <- crimes_clean %>% filter(!is.na(Latitude) & !is.na(Longitude)) # we go from 236549 obser

crimes_sf <- crimes_clean %>%
  st_as_sf(coords = c("Longitude", "Latitude"), crs = 4326, remove = FALSE)

```

Ridership:

```

cta_rail_stations_sf <- cta_rail_stations %>%
  st_as_sf(wkt = "the_geom", crs = 4326)

cta_rail_stations_sf <- cta_rail_stations_sf %>%
  st_filter(chicago_union, .predicate = st_within)

```

Police stations:

```

police_station_clean <- police_station_raw %>% select(-LOCATION)

police_station_clean <- police_station_clean %>%
  mutate(LATITUDE = as.numeric(LATITUDE), LONGITUDE = as.numeric(LONGITUDE))

police_station_clean <- police_station_clean %>% filter(!is.na(LATITUDE) & !is.na(LONGITUDE)) # we go f

police_station_sf <- police_station_clean %>%
  st_as_sf(coords = c("LONGITUDE", "LATITUDE"), crs = 4326, remove = FALSE)

```

Transform to all use same CRS.

```

com_areas <- st_transform(com_areas, CRS.new)
crimes_sf <- st_transform(crimes_sf, CRS.new)
cta_rail_stations_sf <- st_transform(cta_rail_stations_sf, CRS.new)
police_station_sf <- st_transform(police_station_sf, CRS.new)

```

```

cta_rail_stations_sf <- cta_rail_stations_sf %>%
  left_join(cta_ridership_clean,
    by = c("STATION_ID" = "RIDERSHIP_ID"))

```

ESDA

First, plotting police stations + CTA rail stations.

```

tmap_mode("plot")

tm_shape(com_areas) + tm_borders() +
  tm_shape(police_station_sf) + tm_symbols(fill = "blue", shape = 24, size = 0.5) +
  tm_shape(cta_rail_stations_sf) + tm_symbols(fill = "red", size = 0.5) +

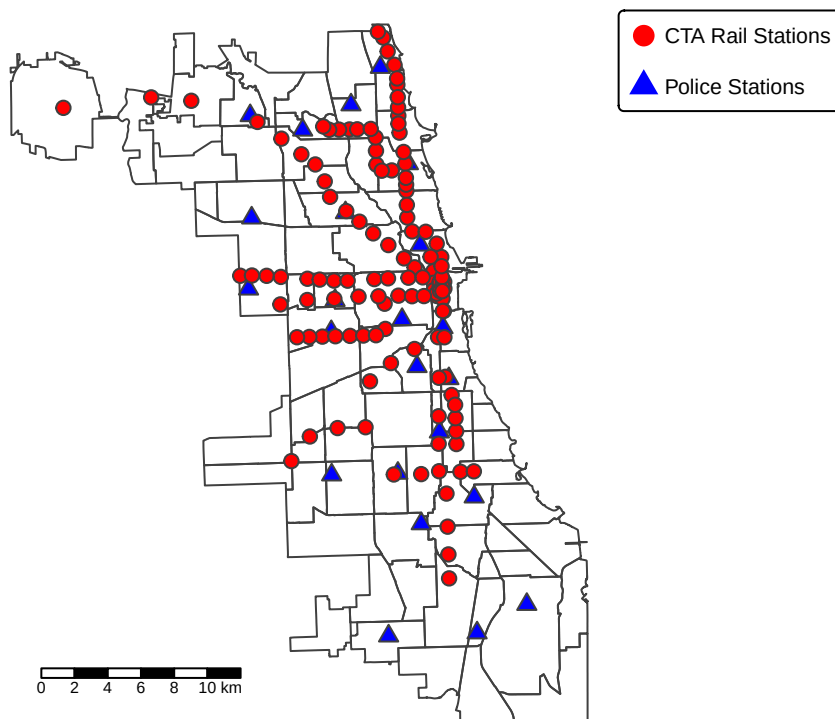
```

```

tm_layout(
  legend.outside = TRUE,
  legend.outside.position = "right",
  outer.margins = c(0.02, 0.02, 0.02, 0.18)) +
tm_layout(
  main.title = paste0("Resource Points"),
  main.title.size = 1.5,
  main.title.position = c("center", "top"),
  legend.outside = TRUE,
  bg.color = "white",
  frame = FALSE
) +
tm_scalebar(position = c("left", "bottom"),
  text.size = 0.5 ) +
tm_add_legend(
  type = "symbol",
  labels = "CTA Rail Stations",
  col = "red",
  shape = 16,
  size = 0.75
) +
tm_add_legend(
  type = "symbol",
  labels = "Police Stations",
  col = "blue",
  shape = 17,
  size = 0.75
)

```

Resource Points



Buffers:

```
buffer_quarter_mi <- st_buffer(cta_rail_stations_sf, 0.25 * 5280)
buffer_quarter_mi <- st_transform(buffer_quarter_mi, CRS.new)

buffer_half_mi <- st_buffer(cta_rail_stations_sf, 0.5 * 5280)
buffer_half_mi <- st_transform(buffer_half_mi, CRS.new)

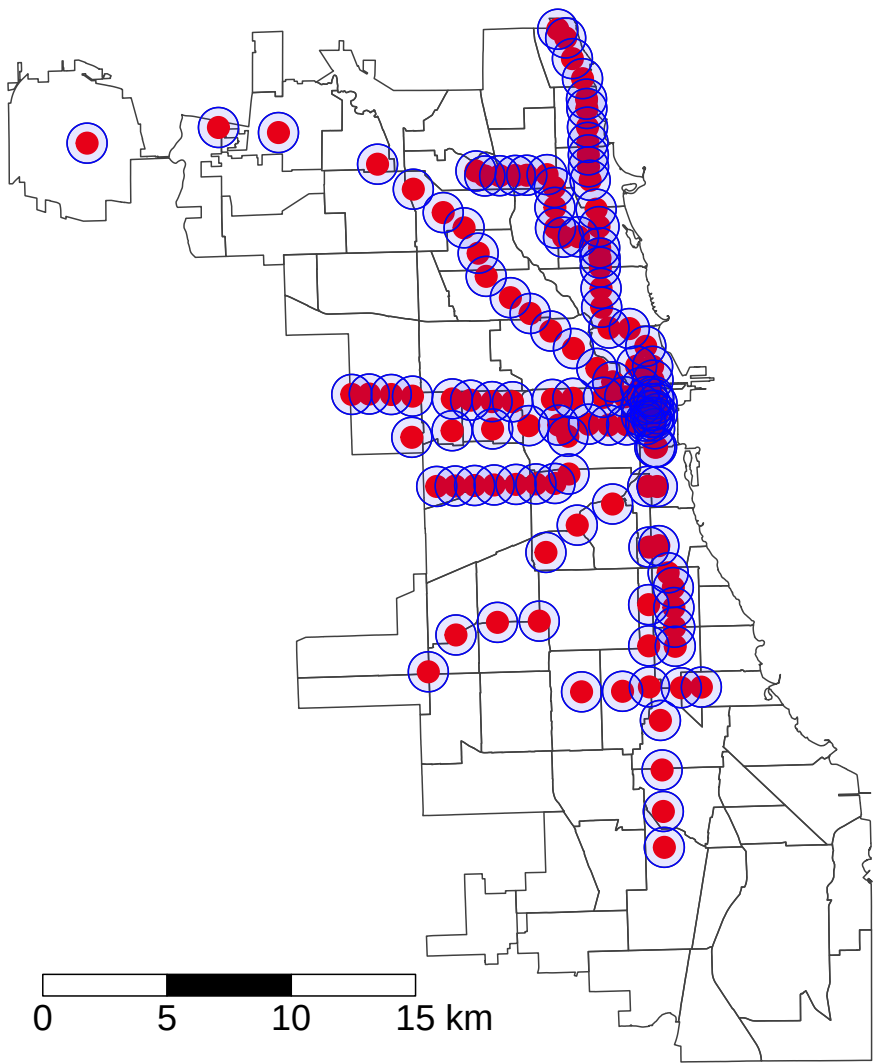
buffer_1_mi <- st_buffer(cta_rail_stations_sf, 5280)
buffer_1_mi <- st_transform(buffer_1_mi, CRS.new)
```

Buffer plots:

```
tmap_mode("plot")
tm_shape(com_areas) +
  tm_borders() +
  tm_shape(cta_rail_stations_sf) +
  tm_dots(fill="red", size=1) +
  tm_shape(buffer_half_mi) +
  tm_fill(col = "blue", alpha = 0.1) +
  tm_borders(col = "blue") +
  tm_layout(
    main.title = paste0("CTA Rail Stations: 0.25-mile Buffer"),
    main.title.size = 1.5,
    main.title.position = c("center", "top"),
    legend.outside = TRUE,
```

```
    bg.color = "white",  
    frame = FALSE  
) +  
tm_scalebar(position = c("left", "bottom"),  
  text.size = 1.5 )
```

CTA Rail Stations: 0.25-mile Buffer



Note: Do we want to get rid of the CTA Rail stations that are part of the network but technically outside of Chicago?

Andy - I think we definitely could, but I'm not exactly sure how to do that in R. Edit - I fixed it above

Crime count in buffer:

```
crime_025_buffer <- lengths(st_intersects(buffer_quarter_mi, crimes_sf))
crime_05_buffer <- lengths(st_intersects(buffer_half_mi, crimes_sf))
crime_1_buffer <- lengths(st_intersects(buffer_1_mi, crimes_sf))
```

```
cta_rail_stations_sf$crime_count_025 <- crime_025_buffer
cta_rail_stations_sf$crime_count_05 <- crime_05_buffer
cta_rail_stations_sf$crime_count_1 <- crime_1_buffer
```

#IMPORTANT: overlapping buffers count crime twice

Police station count in buffer:

```
police_025_buffer <- lengths(st_intersects(buffer_quarter_mi, police_station_sf))
police_05_buffer <- lengths(st_intersects(buffer_half_mi, police_station_sf))
police_1_buffer <- lengths(st_intersects(buffer_1_mi, police_station_sf))
```

```
cta_rail_stations_sf$police_count_025 <- police_025_buffer
cta_rail_stations_sf$police_count_05 <- police_05_buffer
cta_rail_stations_sf$police_count_1 <- police_1_buffer
```

*# From this, only 1 CTA stations has more than 1 police station within a mile radius
will use 1 mile buffer for police station analysis*

Crime Exposure Visualization:

```
cta_rail_stations_sf <- cta_rail_stations_sf %>%
  mutate(
    crime_level_025 = case_when(
      crime_count_025 >= quantile(crime_count_025, 0.75, na.rm = TRUE) ~ "High",
      crime_count_025 >= quantile(crime_count_025, 0.25, na.rm = TRUE) ~ "Medium",
      TRUE ~ "Low"
    ),
    crime_level_025 = factor(crime_level_025, levels = c("Low", "Medium", "High"))
  )
```

```
exposure_025_buffer <- st_buffer(cta_rail_stations_sf, 0.25 * 5280)
```

```
cta_rail_stations_sf <- cta_rail_stations_sf %>%
  mutate(
    crime_level_05 = case_when(
      crime_count_05 >= quantile(crime_count_05, 0.8, na.rm = TRUE) ~ "Very High",
      crime_count_05 >= quantile(crime_count_05, 0.6, na.rm = TRUE) ~ "High",
      crime_count_05 >= quantile(crime_count_05, 0.4, na.rm = TRUE) ~ "Moderate",
      crime_count_05 >= quantile(crime_count_05, 0.2, na.rm = TRUE) ~ "Low",
      crime_count_05 >= quantile(crime_count_05, 0, na.rm = TRUE) ~ "Very Low",
    ),
    crime_level_05 = factor(crime_level_05, levels = c("Very High", "High", "Moderate", "Low", "Very Low"))
  )
```

```

)

exposure_05_buffer <- st_buffer(cta_rail_stations_sf, 0.5 * 5280)

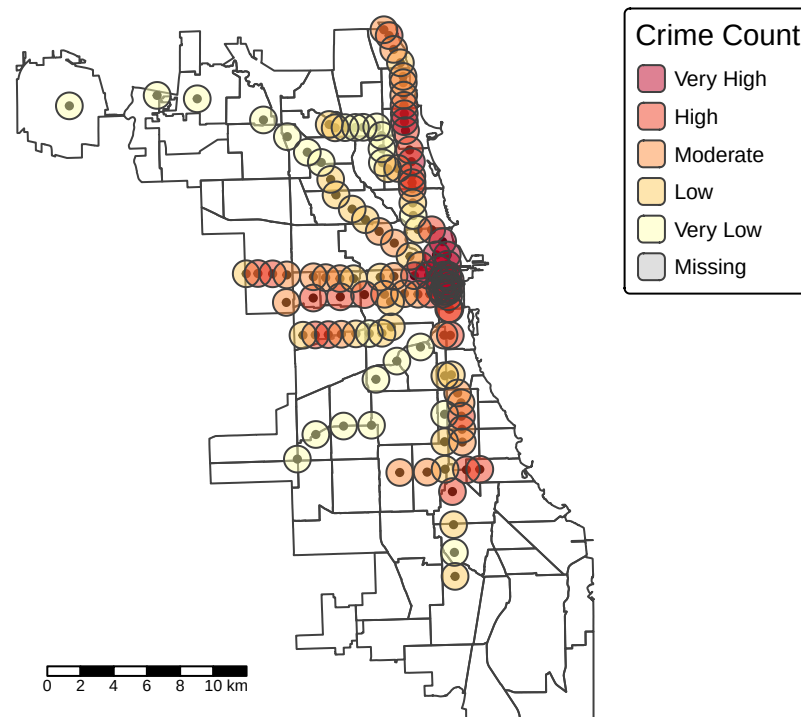
sum(is.na(cta_rail_stations_sf$crime_level_05))

## [1] 0

tmap_mode("plot")
tm_shape(com_areas) +
  tm_borders() +
  tm_shape(cta_rail_stations_sf) +
  tm_dots(fill="black", size=0.25) +
  tm_shape(exposure_05_buffer) +
  tm_fill(
    col = "crime_level_05",
    palette = rev(brewer.pal(5, "YlOrRd")),
    title = "Crime Count",
    border.col = NA,
    alpha = 0.5
  ) +
  tm_layout(
    main.title = paste0("CTA Rail Stations Crime Count: 0.5-mile Buffer"),
    main.title.size = 1.5,
    main.title.position = c("center", "top"),
    legend.outside = TRUE,
    bg.color = "white",
    frame = FALSE
  ) +
  tm_scalebar(position = c("left", "bottom"),
    text.size = 0.5 )

```

CTA Rail Stations Crime Count: 0.5-mile Buffer



```
cta_rail_stations_sf <- cta_rail_stations_sf %>%
  mutate(
    # rate per 10,000 riders (avoid divide-by-zero)
    crime_rate_05 = ifelse(total_rides_2025 > 0,
                          (crime_count_05 / total_rides_2025) * 10000,
                          NA_real_),

    # 5-quantile bins (20% each)
    crime_rate_level_05 = cut(
      crime_rate_05,
      breaks = quantile(crime_rate_05, probs = seq(0, 1, 0.2), na.rm = TRUE),
      include.lowest = TRUE,
      labels = c("Very Low", "Low", "Moderate", "High", "Very High"),
    ),

    crime_rate_level_05 = factor(
      crime_rate_level_05,
      levels = c("Very High", "High", "Moderate", "Low", "Very Low"),
      ordered = TRUE
    )
  )

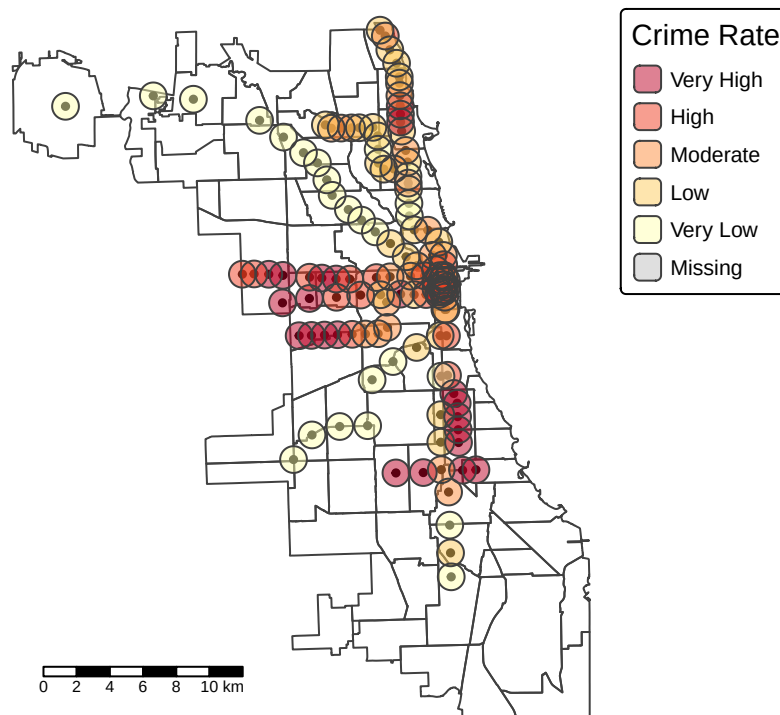
exposure_05_buffer <- st_buffer(cta_rail_stations_sf, 0.5 * 5280)
```

```

tmap_mode("plot")
tm_shape(com_areas) +
  tm_borders() +
  tm_shape(cta_rail_stations_sf) +
  tm_dots(fill="black", size=0.25) +
  tm_shape(exposure_05_buffer) +
  tm_fill(
    col = "crime_rate_level_05",
    palette = rev(brewer.pal(5, "YlOrRd")),
    title = "Crime Rate",
    border.col = NA,
    alpha = 0.5
  ) +
  tm_layout(
    main.title = paste0("CTA Rail Stations Crime Rate: 0.5-mile Buffer"),
    main.title.size = 1.5,
    main.title.position = c("center", "top"),
    legend.outside = TRUE,
    bg.color = "white",
    frame = FALSE
  ) +
  tm_scalebar(position = c("left", "bottom"),
    text.size = 0.5 )

```

CTA Rail Stations Crime Rate: 0.5-mile Buffer



Filtering for Police Stations within a 1 mile radius

```
cta_with_police <- cta_rail_stations_sf %>%
  filter(police_count_1 >= 1)

exposure_05_buffer_police <- st_buffer(cta_with_police, 0.5 * 5280)

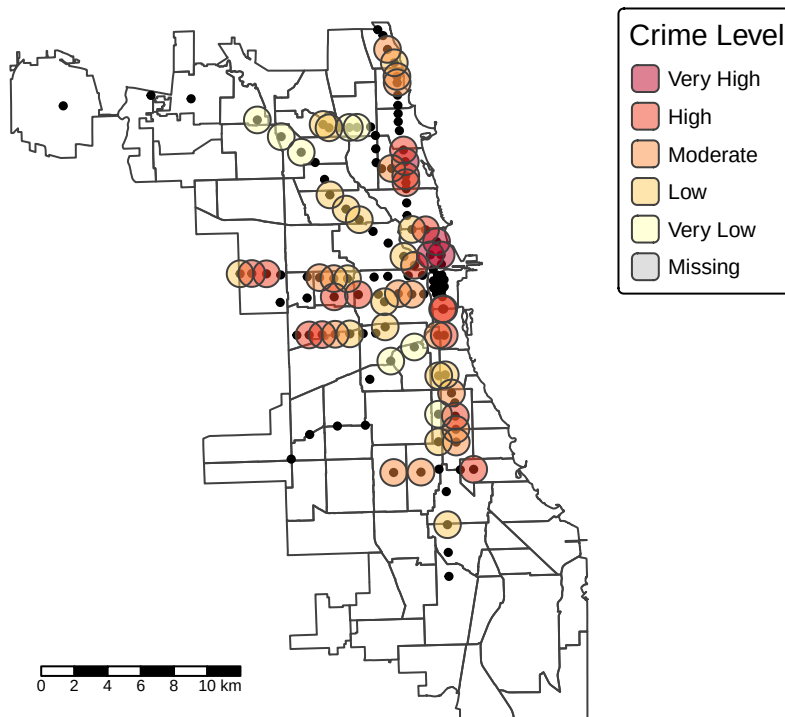
cta_wo_police <- cta_rail_stations_sf %>%
  filter(police_count_1 == 0)

exposure_05_buffer_no_police <- st_buffer(cta_wo_police, 0.5 * 5280)
```

Crime exposure map filtered for police stations nearby:

```
tmap_mode("plot")
tm_shape(com_areas) +
  tm_borders() +
  tm_shape(cta_rail_stations_sf) +
  tm_dots(fill="black", size=0.25) +
  tm_shape(exposure_05_buffer_police) +
  tm_fill(
    col = "crime_level_05",
    palette = rev(brewer.pal(5, "YlOrRd")),
    title = "Crime Level",
    border.col = NA,
    alpha = 0.5
  ) +
  tm_layout(
    main.title = paste0("CTA Rail Stations Crime Exposure: 0.5-mile Buffer with a Police Station Nearby"),
    main.title.size = 1.5,
    main.title.position = c("center", "top"),
    legend.outside = TRUE,
    bg.color = "white",
    frame = FALSE
  ) +
  tm_scalebar(position = c("left", "bottom"),
    text.size = 0.5 )
```

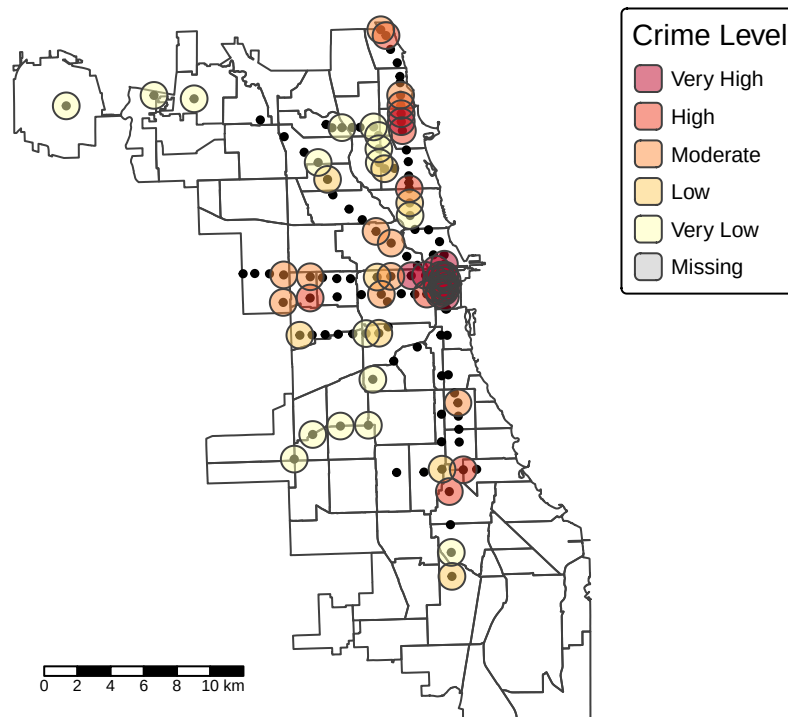
CTA Rail Stations Crime Exposure: 0.5-mile Buffer with a Police Station Nearby



Crime Exposure Map w/o police stations nearby:

```
tmap_mode("plot")
tm_shape(com_areas) +
  tm_borders() +
  tm_shape(cta_rail_stations_sf) +
  tm_dots(fill="black", size=0.25) +
  tm_shape(exposure_05_buffer_no_police) +
  tm_fill(
    col = "crime_level_05",
    palette = rev(brewer.pal(5, "YlOrRd")),
    title = "Crime Level",
    border.col = NA,
    alpha = 0.5
  ) +
  tm_layout(
    main.title = paste0("CTA Rail Stations Crime Exposure: 0.5-mile Buffer without a Police Station Near"),
    main.title.size = 1.5,
    main.title.position = c("center", "top"),
    legend.outside = TRUE,
    bg.color = "white",
    frame = FALSE
  ) +
  tm_scalebar(position = c("left", "bottom"),
    text.size = 0.5 )
```

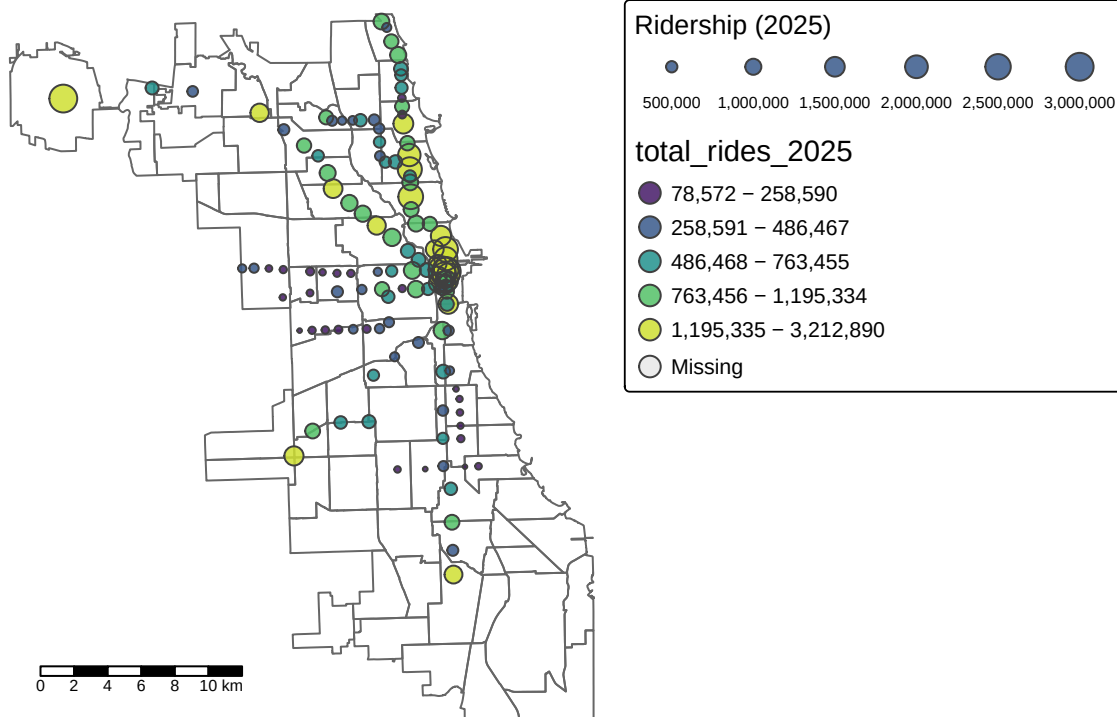
CTA Rail Stations Crime Exposure: 0.5-mile Buffer without a Police Station Nearby



Next, we want to go into ridership levels. Idea: also colorcode like above, with varying point colors.

```
tm_shape(com_areas) +
  tm_borders(col = "grey40") +
  tm_shape(cta_rail_stations_sf) +
  tm_symbols(
    size = "total_rides_2025",
    col = "total_rides_2025",
    palette = "viridis",
    style = "quantile",
    alpha = 0.85,
    title.size = "Ridership (2025)",
    title.col = "Ridership (2025)"
  ) +
  tm_layout(
    main.title = "CTA Rail Station Ridership (2025)",
    legend.outside = TRUE,
    frame = FALSE
  ) +
  tm_scale_bar(position = c("left", "bottom"))
```

CTA Rail Station Ridership (2025)



```
nearest_police_idx <- st_nearest_feature(cta_rail_stations_sf, police_station_sf)
```

```
# compute minimum distance
```

```
min_dist <- st_distance(
  cta_rail_stations_sf,
  police_station_sf[nearest_police_idx, ],
  by_element = TRUE
)
```

```
cta_rail_stations_sf <- cta_rail_stations_sf %>%
  mutate(
    min_dist_police = as.numeric(min_dist),
    min_dist_police_miles = min_dist_police / 5280
  )
```

```
tmap_mode("plot")
```

```
tm_shape(com_areas) +
  tm_borders() +
tm_shape(cta_rail_stations_sf) +
  tm_dots(
    fill = "min_dist_police_miles", # <-- map fill, not col
    fill.scale = tm_scale_intervals(
      n = 5,
      style = "quantile",
      values = "-Oranges"
    )
  )
```

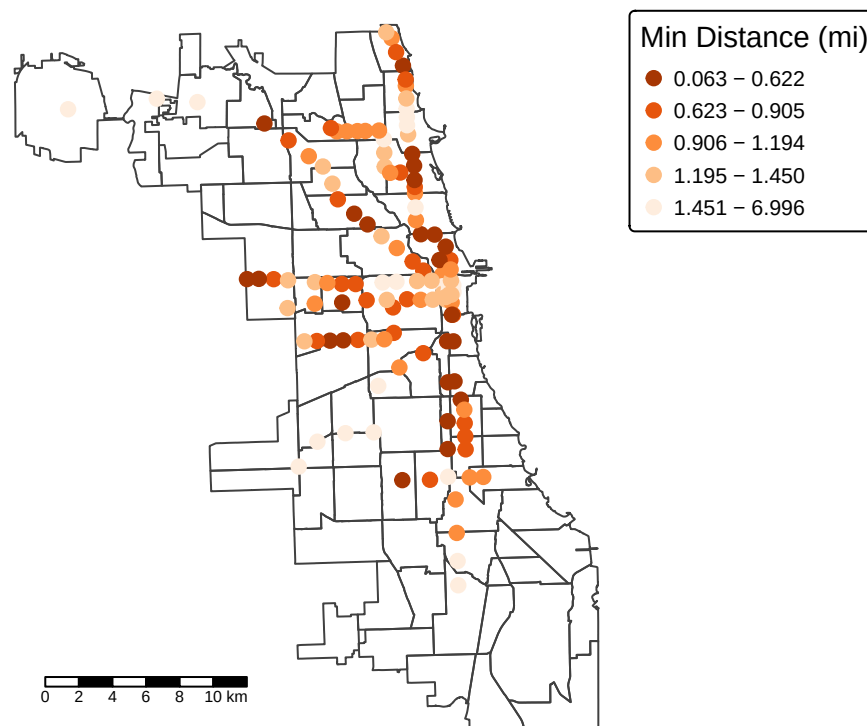


```

),
  fill.legend = tm_legend(title = "Min Distance (mi)",
    size = 0.5
) +
tm_layout(
  main.title = "CTA Stations: Distance to Nearest Police Station",
  legend.outside = TRUE,
  frame = FALSE
) +
tm_scalebar(position = c("left", "bottom"), text.size = 0.5)

```

CTA Stations: Distance to Nearest Police Station



Non-Spatial EDA

Regression with Crime Count 05 and 2025 Ridership

```

plot(cta_rail_stations_sf$crime_count_05,
  cta_rail_stations_sf$total_rides_2025,
  pch = 19,
  col = "darkgray",
  xlab = "Crime Count within 0.5 Miles",
  ylab = "Ridership (2025)",
  main = "CTA 2025 Ridership vs Crime Count")
model <- lm(total_rides_2025 ~ crime_count_05, data = cta_rail_stations_sf)
abline(model, col = "blue", lwd = 2)

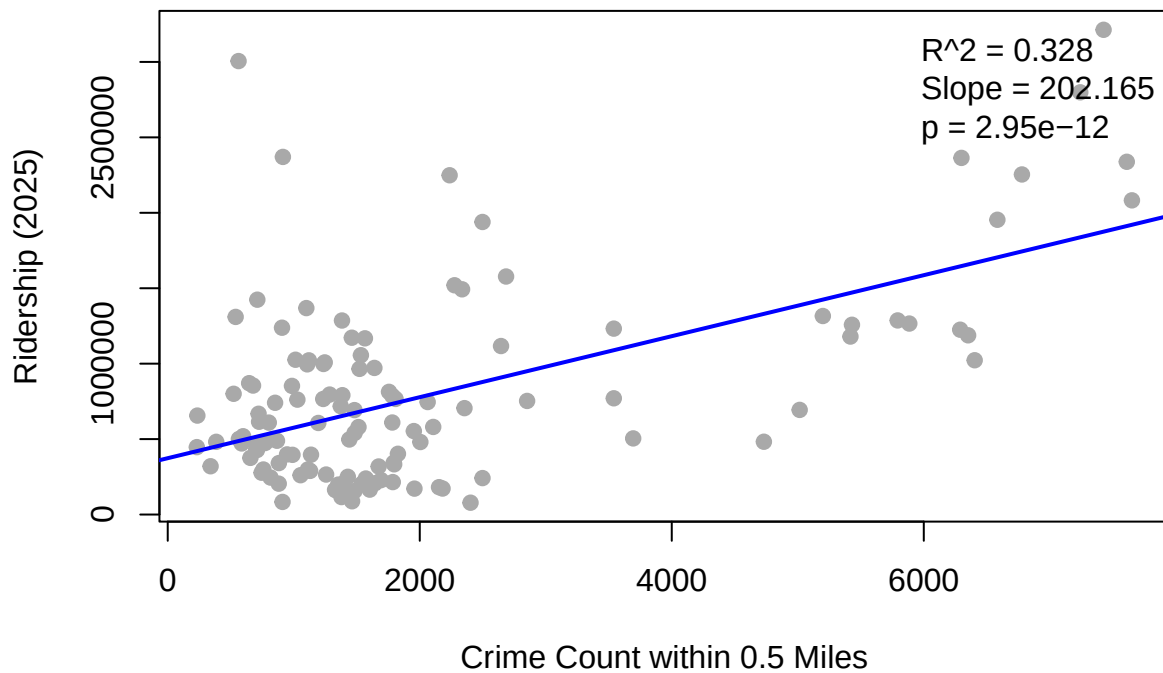
```

```

model_sum = summary(model)
r2 = model_sum$r.squared
slope <- model_sum$coefficients[2, 1]
p_value <- model_sum$coefficients[2, 4]
legend("topright",
      legend = c(
        paste0("R^2 = ", round(r2, 3)),
        paste0("Slope = ", round(slope, 3)),
        paste0("p = ", signif(p_value, 3))
      ),
      bty = "n")

```

CTA 2025 Ridership vs Crime Count



Scatter Plot for Minimum distance vs Crime Count

```

plot(cta_rail_stations_sf$min_dist_police_miles,
     cta_rail_stations_sf$crime_rate_05,
     pch = 19,
     col = "darkgray",
     xlab = "CTA Station Distance from Nearest Police Station",
     ylab = "Crime Rate within 0.5 miles",
     main = "Crime Rate vs. Police Station Minimum Distance")
model <- lm(crime_rate_05 ~ min_dist_police_miles, data = cta_rail_stations_sf)
abline(model, col = "blue", lwd = 2)
model_sum = summary(model)
r2 = model_sum$r.squared
slope <- model_sum$coefficients[2, 1]

```

```

p_value <- model_sum$coefficients[2, 4]
legend("topright",
      legend = c(
        paste0("R^2 = ", round(r2, 3)),
        paste0("Slope = ", round(slope, 3)),
        paste0("p = ", signif(p_value, 3))
      ),
      bty = "n")

```

Crime Rate vs. Police Station Minimum Distance

